

E-COMMERCE SALES ANALYSIS

```
CREATE TABLE superstore (  
    Row_ID SERIAL PRIMARY KEY,  
    Order_ID VARCHAR(50) NOT NULL,  
    Order_Date DATE NOT NULL,  
    Ship_Date DATE NOT NULL,  
    Ship_Mode VARCHAR(50),  
    Customer_ID VARCHAR(50) NOT NULL,  
    Customer_Name VARCHAR(100),  
    Segment VARCHAR(50),  
    Country VARCHAR(50),  
    City VARCHAR(100),  
    State VARCHAR(100),  
    Postal_Code VARCHAR(20),  
    Region VARCHAR(50),  
    Product_ID VARCHAR(50) NOT NULL,  
    Category VARCHAR(50),  
    Sub_Category VARCHAR(50),  
    Product_Name VARCHAR(150),  
    Sales DECIMAL(10, 2) -- Assuming sales as a decimal value with two decimal places  
);
```

```
SELECT * FROM superstore;
```

-- Q1) What percentage of total orders were shipped on the same date?

```
SELECT  
    ROUND((COUNT(DISTINCT Order_ID) / (SELECT COUNT(DISTINCT Order_ID) AS total_orders FROM superstore)) *  
    100, 2) AS Same_Day_Shipping_Percentage  
FROM  
    superstore  
WHERE  
    Order_Date = Ship_Date;
```

-- Q2) Name top 3 customers with highest total value of orders?

```
SELECT
    Customer_Name,
    ROUND(SUM(sales), 3) AS TotalOrderValue
FROM
    superstore
GROUP BY
    Customer_Name
ORDER BY
    SUM(sales) DESC
LIMIT 3;
```

-- Q3) Find the top 5 items with the highest average sales per day?

```
SELECT Product_ID,
    ROUND(AVG(sales), 3) AS Average_Sales FROM superstore
GROUP BY
    Product_ID
ORDER BY
    Average_Sales DESC
LIMIT 5;
```

-- Q4) Write a query to find the average order value for each customer, and rank the customers by their average order value?

```
SELECT
    Customer_Name,
    ROUND(AVG(sales), 3) AS avg_order_value,
    DENSE_RANK() OVER (ORDER BY AVG(sales) DESC) AS sales_rank
FROM
    superstore
GROUP BY
    Customer_Name;
```

-- Q5) Give the name of customers who ordered highest and lowest orders from each city?

WITH cte AS (

SELECT

City,

ROUND(MAX(sales), 4) AS highest_order,

ROUND(MIN(sales), 4) AS lowest_order

FROM superstore GROUP BY City

),

highest_orders AS (

SELECT s.City, cte.highest_order, cte.lowest_order, s.Customer_Name FROM superstore s

INNER JOIN

cte ON s.City = cte.City

WHERE s.Sales = cte.highest_order

),

lowest_orders AS (

SELECT

s.City,

cte.highest_order, cte.lowest_order, s.Customer_Name

FROM superstore s

INNER JOIN

cte ON s.City = cte.City

WHERE

s.Sales = cte.lowest_order

)

SELECT

h.City,

h.highest_order,

h.Customer_Name AS highest_order_customer,

l.lowest_order,

l.Customer_Name AS lowest_order_customer

FROM highest_orders h

INNER JOIN

lowest_orders l ON h.City = l.City ORDER BY h.City;

-- Q6) What is the most demanded sub-category in the west region?

```
SELECT
    Sub_Category,
    ROUND(SUM(sales), 3) AS total_quantity
FROM
    superstore
WHERE
    Region = 'West'
GROUP BY
    Sub_Category
ORDER BY
    total_quantity DESC
LIMIT 1;
```

-- Q7) Which order has the highest number of items?

```
SELECT
    order_id,
    COUNT(order_id) AS num_item
FROM
    superstore
GROUP BY
    order_id
ORDER BY
    num_item DESC
LIMIT 1;
```

-- Q8) Which order has the highest cumulative value?

```
SELECT
    order_id,
    ROUND(SUM(sales), 3) AS order_value FROM superstore
GROUP BY order_id ORDER BY order_value DESC LIMIT 1;
```

-- Q9) Which segment's order is more likely to be shipped via first class?

```
SELECT
    segment,
    COUNT(order_id) AS num_of_ordr FROM superstore
WHERE
    ship_mode = 'First Class'
GROUP BY segment
ORDER BY num_of_ordr DESC;
```

-- Q10) Which city is least contributing to total revenue?

```
SELECT
    city,
    ROUND(SUM(sales), 3) AS TotalSales
FROM superstore
GROUP BY
    city
ORDER BY
    TotalSales ASC
LIMIT 1;
```

-- Q11) What is the average time for orders to get shipped after order is placed?

```
/*
SELECT
    AVG(DATEDIFF(ship_date, order_date)) AS avg_ship_time
FROM
    superstore;
*/
SELECT
    AVG(AGE(ship_date, order_date)) AS avg_ship_time
FROM
    superstore;
```

**/* Q12) Which segment places the highest number of orders from each state
and which segment places the largest individual orders from each state? */**

```
WITH cte AS (  
    SELECT state, segment,  
           COUNT(order_id) AS num_orders,  
           RANK() OVER (PARTITION BY state ORDER BY COUNT(order_id) DESC) AS state_rank  
    FROM superstore GROUP BY state,segment  
)  
  
SELECT state,segment FROM cte WHERE state_rank = 1;
```

-- Q13) Find the maximum number of days for which total sales on each day kept rising?

```
WITH sales_sequence AS (  
    SELECT  
        Order_Date,  
        SUM(Sales) AS TotalSales,  
        ROW_NUMBER() OVER (ORDER BY Order_Date) AS rn  
    FROM  
        superstore  
    GROUP BY  
        Order_Date  
)  
  
rising_days AS (  
    SELECT  
        s1.Order_Date,  
        COUNT(*) AS rising_day_count  
    FROM sales_sequence s1  
    INNER JOIN  
        sales_sequence s2 ON s1.TotalSales < s2.TotalSales AND s1.rn < s2.rn  
    GROUP BY  
        s1.Order_Date  
)  
  
SELECT  
    MAX(rising_day_count) AS max_rising_days  
FROM rising_days;
```